MTH309 PRACTICE PROBLEMS WEEK 7

Question 1. Construct a probability space $(\Omega, \mathcal{F}, \mu)$ and a real valued integrable function X on this space such that X is not bounded.

Question 2. Let $f : (\Omega, \mathcal{F}) \to (\mathbb{R}, \mathbb{B}_{\mathbb{R}})$ be a measurable function and let μ be a σ-finite measure on $(\Omega, \mathcal{F})$. Recall that f^+ and f^- denote the positive and negative parts of f respectively.

(a) Define $\nu^+(A) := \int_A f^+ \, d\mu, \forall A \in \mathcal{F}$ and $\nu^-(A) := \int_A f^- \, d\mu, \forall A \in \mathcal{F}$. Are ν^+ and ν^- σ-finite measures on $(\Omega, \mathcal{F})$?

(b) Suppose f is integrable with respect to the measure μ. Are ν^+ and ν^- finite measures on $(\Omega, \mathcal{F})$?

(c) Suppose f is integrable with respect to the measure μ. Consider the set function $\nu(A) := \nu^+(A) - \nu^-(A), \forall A \in \mathcal{F}$. Is the domain of ν the σ-field $\mathcal{F}$? (i.e., is ν defined for all sets $A \in \mathcal{F}$?). Is ν countably additive on its domain, i.e. given a sequence of pairwise disjoint sets $\{A_n\}$ from its domain, is $\nu\left(\cup_{n=1}^{\infty} A_n\right) = \sum_{n=1}^{\infty} \nu(A_n)$?

Question 3. Let $\{X_n\}_n$ be a sequence of non-negative, integrable random variables on $(\Omega, \mathcal{F}, \mathbb{P})$. Show that

$$\mathbb{E}\left(\sum_n X_n\right) = \sum_n \mathbb{E}X_n.$$

Does the result hold if $\mathbb{E}X_n = \infty$ for some n?

Question 4. Let $\mu_1, \mu_2, \cdots$ be a sequence of measures on a measurable space $(\Omega, \mathcal{F})$ and let $\alpha_1, \alpha_2, \cdots$ be a sequence of non-negative real numbers. Given any measurable function $f : (\Omega, \mathcal{F}) \to (\mathbb{R}, \mathbb{B}_{\mathbb{R}})$, show that

$$\int f \, d(\sum_n \alpha_n \mu_n) = \sum_n \alpha_n \int f \, d\mu_n,$$

provided the integrals exist.

Question 5. Let $f : \mathbb{R} \to \mathbb{R}$ be Borel measurable and fix $a \in \mathbb{R}$. Consider the two integrals '$\int_{\mathbb{R}} f(x) \, dx$' and '$\int_{\mathbb{R}} f(x-a) \, dx$' with respect to the Lebesgue measure. If one integral exists, show the existence of the other. In this case, show that the two integrals are actually equal.

Question 6. If two Borel measurable functions f and g are both μ-integrable, show that $f + g$ and $f - g$ are also μ-integrable.